

Yufeng Gu

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Education

University of Michigan

Ph.D., Computer Science and Engineering Department

Ann Arbor, MI, USA

Sep. 2020 - Apr. 2026

Zhejiang University (ZJU)

B.Eng., College of Information Science and Electronic Engineering

Hangzhou, China

Sep. 2016 - June 2020

Experience

Meta Platform Inc.

Research Scientist, AI system Co-design

Menlo Park, CA, USA

May 2026 - Now

Conduct research on hardware performance modeling and optimization on large-scale GenAI training and inference on the GPU cluster and In-house AI accelerator.

University of Michigan

Graduate Research Assistant, Advisor: Reetuparna Das

Ann Arbor, MI, USA

Sep. 2020 - Apr. 2026

Co-design architecture and system for large-scale emerging applications, including Generative AI and genome sequencing. Publish papers at top conferences including ISCA, ASPLOS and HPCA, awarded as Communication of ACM Research Highlights (24/10,000+) and HPCA Distinguished Artifact Honorable Mention (3/29).

Tenstorrent Inc.

Performance Architect Intern, Manager: Wei-han Lien

Santa Clara, CA, USA

May 2023 - Aug. 2023

Build a profile-guided agile performance model for Tensix AI accelerator, including computation and communication between RISC-V cores, with <10% error rate. This model reduces the performance modeling time from a few hours to ~5 minutes, enabling fast kernel development and optimization.

Intel Labs

Graduate Research Intern, Manager: Nilesh Jain

Hillsboro, OR, USA

June 2022 - Aug. 2022

Develop an offline profile and online prediction-based resource scheduler for Quality of Service (QoS)-aware job co-location for ML inference, achieving up to 1.4× request throughput improvement over single model deployment and reducing scheduling times from ~30 minutes to ~10 seconds compared to state-of-the-art schedulers.

Yale University

Undergraduate Research Intern, Advisor: James Duncan, Xiaoxiao Li

New Heaven, CT, USA

Nov. 2019 - Mar. 2020

Interpretation on ASD with fMRI and Deep Learning Models. Publications on MICCAI and MedIA.

École Polytechnique Fédérale De Lausanne (EPFL)

Undergraduate Research Intern, Advisor: Babak Falsafi

Lausanne, Switzerland

July 2019 - Sep. 2019

Patched QFlex Simulator (from Makefile to CMake compilation) and accelerated activation functions for DNN.

Award & Honors

Outstanding Young Speaker Award (APPT 2025)

Hardware-Software Co-Design for Scalable LLM Inference

July 2025

Distinguished Artifact Honorable Mention (HPCA 2025)

Multi-Dimensional Vector ISA Extension for Mobile In-Cache Computing. (3/29 Artifacts)

March 2025

Communication of ACM (CACM) Research Highlights

GenDP: A Framework of Dynamic Programming Acceleration for Genome Sequencing Analysis.

July 2024

CACM Research Highlights section selects 24/10,000+ papers from ACM conferences per year, reprinting the most significant and influential results across Computer Science.

University of Michigan Rackham Graduate Student Research Grant (\$3000, role: PI)	<i>Apr. 2024</i>
Pangenomics Benchmark Suite and Characterization.	
University of Michigan Rackham Travel Grant, ISCA/HPCA Travel Grant	<i>2023, 2025</i>
Fellowship of Summer@EPFL (2% applicants awarded)	<i>July 2019</i>
Tang Lixin Fellowship (2% students awarded)	<i>Nov. 2017, 2018, 2019</i>
Outstanding Student Leaders in Zhejiang University (3% students awarded)	<i>Oct. 2017, 2019</i>
First-Class Scholarship for Outstanding Students (2% students awarded)	<i>Oct. 2017</i>

Under-Review Publications

Yufeng Gu, Reetuparna Das. "Adaptive Scheduling for Large Language Model Inference on Heterogeneous Server Farm" (*Under Submission to ASPLOS'27*)

Yufeng Gu, Samel Gobriel, Nilesh Jain, Reetuparna Das. "BRIDGE: A Systolic Processing-In-Memory Architecture for Narrow GEMMs in LLM Inference" (*Under Submission to IEEE Transactions on Computers*)

Yufeng Gu, Vui Seng Chua, Nilesh Jain, Ravishankar Iyer, Reetuparna Das. "Fast Resource Management for Quality of Service-Aware Co-Location of Machine Learning Inference." (*Under Submission to SC'2026*)

Sumanth Umesh, Ning Liang, **Yufeng Gu**, Reetuparna Das. "Addressing In-Memory Database Bottlenecks with CXL Memory Expansion." (*Under Submission to VLDB*)

Peer-Reviewed Publications

* equal contribution

[10] Jiajun Luo, Yu Jin, **Yufeng Gu**, Jinyi Deng, Yang Hu and Shuwen Deng. "Stone-Skimming: Attacking Big-Data Analytics Applications with Page Table Walk." To appear in 63rd ACM/IEEE Design Automation Conference (DAC'26).

[9] Noah Kaplan, Jan-Niklas Schmelzle, **Yufeng Gu**, Christopher Batten, Reetuparna Das. "PangenomicsBench: A Benchmark Suite and Characterization of Pangenomics." In IEEE International Symposium on Workload Characterization (IISWC'25).

[8] **Yufeng Gu**, Arun Subramaniyan, Tim Dunn, Alireza Khadem, Kuan-Yu Chen, Somnath Paul, Md Vasimuddin, Sanchit Misra, David Blaauw, Satish Narayanasamy, Reetuparna Das. "GenDP: A Framework of Dynamic Programming Acceleration for Genome Sequencing Analysis." Communications of the ACM, 2025.

[7] Alireza Khadem*, Kamalavasan Kamalakkannan*, Zhenyan Zhu, Akash Poptani, **Yufeng Gu**, Jered Benjamin Dominguez-Trujillo, Nishil Talati, Daichi Fujiki, Scott Mahlke, Galen Shipman, Reetuparna Das. "DX100: Programmable Data Access Accelerator for Indirection." In Proceedings of the 52th Annual International Symposium on Computer Architecture (ISCA'25).

[6] **Yufeng Gu***, Alireza Khadem*, Sumanth Umesh, Ning Liang, Xavier Servot, Onur Mutlu, Ravishankar Iyer, Reetuparna Das. "PIM Is All You Need: A CXL-Enabled GPU-Free System for Large Language Model Inference." In Proceedings of the 30th ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS'25). **In Progress of Technology Transfer to SK Hynix's Next Generation Datacenter**

[5] Alireza Khadem, Daichi Fujiki, Hilbert Chen, **Yufeng Gu**, Nishil Talati, Scott Mahlke, Reetuparna Das. "Multi-Dimensional Vector ISA Extension for Mobile In-Cache Computing." In 2025 IEEE International Symposium on High-Performance Computer Architecture (HPCA'25). **Distinguished Artifact Honorable Mention (3/29)**

[4] **Yufeng Gu**, Arun Subramaniyan, Tim Dunn, Alireza Khadem, Kuan-Yu Chen, Somnath Paul, Md

Vasimuddin, Sanchit Misra, David Blaauw, Satish Narayanasamy, Reetuparna Das. "GenDP: A Framework of Dynamic Programming Acceleration for Genome Sequencing Analysis." In Proceedings of the 50th Annual International Symposium on Computer Architecture (ISCA'23). **Communication of ACM Research Highlights (24/10,000+)**

[3] Arun Subramaniyan, **Yufeng Gu**, Timothy Dunn, Somnath Paul, Md Vasimuddin, Sanchit Misra, David Blaauw, Satish Narayanasamy, and Reetuparna Das. "GenomicsBench: A Benchmark Suite for Genomics." In IEEE International Symposium on Performance Analysis of Systems and Software (ISPASS'21).

[2] Xiaoxiao Li, **Yufeng Gu**, Nicha Dvornek, Lawrence H. Staib, Pamela Ventola, and James S. Duncan. "Multi-site fMRI analysis using privacy-preserving federated learning and domain adaptation: ABIDE results." Medical Image Analysis 65: 101765. $IF = 11.1$

[1] Xiaoxiao Li, Yuan Zhou, Nicha C. Dvornek, **Yufeng Gu**, Pamela Ventola, and James S. Duncan. "Efficient Shapley Explanation for Features Importance Estimation Under Uncertainty." In International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI'20).

Talks

Architecture and System Co-Design for Scalable LLM Inference

Chinese University of Hong Kong (CUHK)

Dec. 2025

PhD Forum at APPT 2025

July 2025

Los Alamos National Laboratory

May 2025

CXL-enabled PIM system for LLM inference

SAFARI Live Seminar (Host: Prof. Onur Mutlu)

June 2025

MCCSys workshop co-located with ASPLOS 2025

Mar. 2025

Genomics Benchmark Suite and Accelerator Design

Computer Architecture Seminar at UCF (Host: Prof. Di Wu)

Mar. 2024

Cornel University (Host: Prof. Christopher Batten)

Feb. 2024

Peisu Xia Forum at ICT, CAS

Dec. 2023

Teaching

Guest Lecture at UMich EECS 570 Parallel Computer Architecture (Fall 2025 and Winter 2026)

Programming at Discovery Engineering program at University of Michigan

July. 2022

Mathematics at Tuanlin Primary School, Guizhou, China

Aug. 2017

Services

Journal Reviewer

- IEEE Transaction on Computer (TC)
- Journal of Systems Architecture (JSA)
- IEEE Transaction on Big Data (TBD)
- IEEE Computer Architecture Letter (CAL)

Artifact Evaluation Committee

- International Symposium on Computer Architecture (ISCA) 2023, 2024, 2025
- IEEE International Symposium on Workload Characterization (IISWC) 2021

Mentorship

Linus Wang B.S. UM → Ph.D. at UMich

Jan. 2025 - Apr. 2026

Wenjie Geng M.S. UM

Sep. 2024 - Aug. 2026

Zesen Zhao M.S. UM → Ph.D. at UMich	<i>Jan. 2025 - Aug. 2025</i>
Mayne Mei B.Eng. UM → Engineer at Etched	<i>Jan. 2024 - Apr. 2024</i>
Yuzhe Ruan B.S. UM → M.S. at Yale	<i>Sep. 2023 - Apr. 2024</i>
Donglin Yu B.Eng. → M.Eng. at UIUC	<i>Jan. 2023 - Apr. 2024</i>
Dawit Melka B.Eng. Addis Ababa University → Engineer at iCog Labs	<i>May 2022 - Aug. 2022</i>
James Gu B.Eng. UM → Engineer at Amazon	<i>May 2022 - Aug. 2022</i>